Power Budget

Team Number:	104
Project Name:	Force Sensing Gripper
Team Member Names:	
Version:	**Gripper Actuation**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators,

All Major Components	**Component Name**	**Part Number**	**yVoltageRar**	**#**	**aximumCurrea**	**alCurrent(mA)**	**Unit**
	Microcontroller (MCU)	default	5V	1	45	45	mA
	Opamp	TLV9002IDR	+5V	1	2	2	mA
	Motor Driver Logic Stage	Cytron MD13S	5V	1	15	15	mA
	DC Brushed Gearmotor	Pololu 25D	+12V	1	2200	2200	mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the

+12V Power Rail	**Component Name**	**Part Number**	**yVoltageRar**	**#**	**aximumCurrea**	**alCurrent(mA)**	**Unit**
	DC Brushed Gearmotor	Pololu 25D	+12V	1	2200	2200	2200
					Subtotal	2200	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	2750	mA
c1. Regulator or Source Ch	Direct Battery Bus	Unregulated W	+12V	1	5000	5000	mA
					Total Remaining Current Available on +12V Rail	2250	mA

+5V Power Rail	**Component Name**	**Part Number**	**yVoltageRar**	**#**	**aximumCurrea**	**alCurrent(mA)**	**Unit**
	Microcontroller (MCU)	Default	5V	1	45	45	mA
	Opamp	TLV9002IDR	5V	1	2	2	mA
	Motor Driver Logic Stage	Cytron MD13S	5V	1	15	15	mA
					Subtotal	62	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	77.5	mA
c2. Regulator or Source Ch	+5V Regulator	LM7805	7V to 35V	1	1000	1000	mA
					Total Remaining Current Available on +5V Rail	922.5	mA

-5V Power Rail	**Component Name**	**Part Number**	**yVoltageRar**	**#**	**aximumCurrea**	**alCurrent(mA)**	**Unit**
						0	mA
						0	mA
					Subtotal	0	mA
					Safety Margin	0%	
					Total Current Required on -5V Rail	0	mA
c3. Regulator or Source Ch	0	0	0	0	0	0	mA
					Total Remaining Current Available on -5V Rail	0	mA

+3.3V Power Rail	**Component Name**	**Part Number**	**yVoltageRar**	**#**	**aximumCurrea**	**alCurrent(mA)**	**Unit**
						0	mA
						0	mA
						0	mA
					Subtotal	0	mA
					Safety Margin	0%	
					Total Current Required on +3.3V Rail	0	mA
c4. Regulator or Source Ch	0	0	0	0	0	0	mA
					Total Remaining Current Available on 3.3V Rail	0	mA

C. For each power rail above, select a specific voltage regulator using the same process as for major component selection.

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the

External Power Source 1	**Component Name**	**Part Number**	**yVoltageRar**	**Outpu**	**aximumCurrea**	**alCurrent(mA)**	**Unit**
Battery Pack	Li-Ion Battery Pack	Rechargeable F	12V Nominal	+24V	5000	5000	mA
Power Rails Connected to External Power Source 1	Direct Actuator Supply	Unregulated Li	+12V	1	2750	2750	mA
	+5V Regulator	LM7805	(range)	1	77.5	77.5	mA
				1	0	0	mA
					Total Remaining Current Available on External Power Source 1	2172.5	mA

External Power Source 2	**Component Name**	**Part Number**	**yVoltageRar**	**Outpu**	**aximumCurrea**	**alCurrent(mA)**	**Unit**
Power Source 2 Selection	Battery	full part numbe	+9V	-9V			mA
Power Rails Connected to External Power Source 2	-5V Regulator	(full part numbe	(range)	1		0	mA
					Total Remaining Current Available on External Power Source 2	0	mA

E. Calculate Battery Life (if applicable). For each battery, also check the worst-case lifetime of the

Component Name	**Part Number**	**yVoltageRange**	**Capacity(mAh)**	**redByRegulators**	
Battery Pack	3S Li-Ion Config	+12V	2200	2262	
			Battery Life	0.972590628	hours

Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit